Mapping the pH heterogeneity of the secretory pathway by guided unmixing of fluorescence lifetime imaging data

(Data Analysis)

Trevanan Gerald Yuan (2026)

Contents

Abstract.....3

1. **Introduction**.....5

1.1 Importance of Identifying Intracellular pH Gradients.....5

1.2 Evolution of pH Biosensors.....6

1.3 Using FLIM for Quantitative Analysis.....7

1.4 Data Analysis of the pH Heterogeneity of the Secretory Pathway.....8

1.5 Project Aims and Hypothesis.....11

2. **Experimental Procedures**.....13

2.1 Description of the FLIM Dataset.....13

2.2 Data Pre-Processing and Optimisation.....14

2.3 Generation of Reference Data for Guided Analysis.....16

2.4 Non-Negative Matrix Factorisation (NMF) Framework.....17

2.5 Guided NMF Analysis of Calibration and Cellular Data.....18

2.6 Calculation and Visualisation.....20

3. **Results**.....22

3.1 Impact of Partial Noise Whitening on Data Quality.....22

3.2 Effect of Spatial Binning on pH Map Quality.....22

3.3 Validation of Guided Analysis Using Recombinant Calibrated Data.....23

3.4 Application of Cellular Probe Data.....24

4. **Discussion**.....28

4.1 Integration of Experimental Findings with Existing Literature.....28

4.2 The Photon Budget and its Implications.....29

4.3 Background Noise: An Ongoing Issue.....30

4.4 The Conceptual Framework of NMF and its Application.....31

4.5 The Pathway Forward: Moving from Supervised Guidance to Unsupervised Discovery.....32

4.6 Conclusion.....34

References.....35

Summary

Introduction: Intracellular pH gradients along the secretory pathway regulate protein folding, glycosylation, sorting and transport, and their disruption is linked to diseases such as Congenital Disorders of Glycosylation (CDG) and lysosomal storage disorders. Fluorescence Lifetime Imaging Microscopy (FLIM) of the genetically encoded biosensor RpHLuorin2 offers a single-excitation, concentration-independent alternative to radiometric pH measurements. However, extracting accurate pH information from FLIM datasets, where the signal at each pixel comprises multiple protonation states and autofluorescence, remains a significant analytical challenge.

Aims: The initial aim of this data analysis project was to develop and critically evaluate an unsupervised non-negative matrix factorization (NMF) framework for mapping the pH of intracellular organelles from FILM data. Because the unsupervised approach did not give interpretable results at the available photon counts, the analysis was moved to a supervised (guided) approach. The accuracy and limitations of the supervised method were tested, and a future route towards integration with next-generation unsupervised algorithms was proposed.

Methodology: Recombinant RpHLuorin2 was calibrated in buffers with known pH values (4.5, 5.5, 6.0, and 7.5) to generate reference fluorescence decay curves. These references were used as fixed components in an NMF algorithm within the Hyperspectral Image Analysis (HIA) software, implementing a guided analysis framework. The framework was validated on recombinant samples, including two pH values (5.0 and 6.5) that were not used as references, and applied to cellular FLIM data. Spatial pH maps were generated from the component concentration maps through abundance-weighted averages calculated in ImageJ and MATLAB. Performance was evaluated on signal-to-noise ratio (SNR), resolution of intermediate pH values, and agreement with expected pH values.

Key Results: The guided unmixing method generated spatially coherent pH maps capable of distinguishing broad pH ranges, with colour-coded that followed the nominal pH of the calibration samples. Recombinant pH 4.5 and 7.5 samples gave histogram peaks at pH 4.7 and 7.3 respectively, indicating a compression of estimates towards the centre of the calibrated range. Several limitations were identified: poor resolution of intermediate pH

values, dependence on spatial binning (4 x 4) to achieve sufficient SNR, variable performance across cellular datasets that were analysed identically, and over-parameterisation of the four-component model, in which the intermediate references contributed minimally to the final analysis. Background noise from overlapping signals further complicated interpretation.

Discussion and Conclusion: The results validate the principle of component-guided FLIM unmixing for determining pH, and define the constraints of this study, namely the photon budget (SNR) and model rigidity. While supervised approaches can generate qualitative pH maps, they are vulnerable to systematic error when the reference used as a guide is not representative of the *in cellulo* photophysics of the sensor. The necessity for spatial binning highlights the photon budget limitations of FLIM experiments. These results motivate a shift from supervised toward unsupervised, model-blind analysis, such as uFLIM and phasor S-FLIM, ideally initialised with guided estimates and calibrated against in-cell references. Such integration could enable quantitative, high-resolution and multi-parameter mapping of organelle pH, and allow the dynamic pH changes experienced by trafficking cargo to be followed.

1. **Introduction**

1.1 Importance of Identifying Intracellular pH Gradients

In eukaryotic cells, internal compartmentalisation creates distinct biochemical environments (Gabaldón and Pittis 2015), and differences in luminal pH between compartments regulate organelle function (Jain and Zoncu 2022). The secretory pathway provides an ideal model for observing pH heterogeneity, as it exhibits a finely tuned pH gradient. A near neutral pH of approximately 7 is observed in the endoplasmic reticulum (ER), the Golgi apparatus acidifies progressively from approximately 6.7 at the cis face to 6.0 at the trans face, and secretory vesicles (pH approximately 5.2) and lysosomes (pH approximately 4.7) are more acidic still environment (Paroutis et al. 2004; Casey et al. 2009; Linders et al. 2022). FILM of RpHLuorin2 targeted to individual compartments reproduced this gradient, giving apparent mean luminal pH values of 7.2 in the ER, 6.1 in the medial-Golgi, 5.9 in the trans-Golgi and 4.7 in the lysosomes (Linders et al. 2022), as seen in Figure 1.

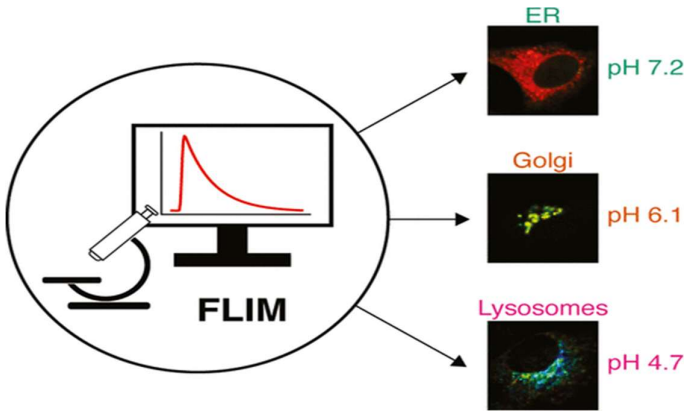


Figure 1. Apparent luminal pH of the ER (7.2), Golgi apparatus (6.1) and lysosomes (4.7) measured by FILM of organelle-targeted RpHLuorin2. Reproduced from Linders et al. (2022).

This pH gradient varies between organelles and is essential for controlling critical cellular processes. By altering the protonation state of amino acid side chains, pH influences protein folding, stability and molecular recognition (Petukh et al. 2013). It also modulates luminal enzyme activity, and controls the binding of cargo to trafficking receptors such as ERGIC-53 and the KDEL receptor, so that cargo is released, retrieved and packaged according to regulatory requirements (Appenzeller-Herzog et al. 2004; Bräuer et al. 2019). Disruptions to

pH homeostasis have been directly linked to disease. Raised Golgi pH, for example, impairs terminal N-glycosylation (Rivinoja et al. 2009), and mutations in subunits of the vacuolar H^+-ATPase cause Congenital Disorders of Glycosylation (CDG) (Kornak et al. 2007; Linders et al. 2020), while defects in pH homeostasis are also found in various lysosomal storage diseases (Linders et al. 2022). Developing methods that measure pH with high spatiotemporal accuracy at the subcellular level is therefore of great importance to furthering our understanding of cell biology and the mechanisms of disease (Casey et al. 2009; Zhang et al. 2024; Linders et al. 2022).

1.2 Evolution of pH Biosensors

Interest in how drugs, mutations and stress alter cellular pH has driven the rapid evolution of genetically encoded fluorescent biosensors, including recent lifetime-based sensors combined with deep learning (Rennick et al. 2022). The earliest genetically encoded sensors were pH-sensitive variants of green fluorescent protein (GFP), the pHLuorins (Miesenböck et al. 1998; Royle et al. 2008). Ecliptic pHLuorin reports pH through fluorescence intensity alone, which makes it susceptible to false readings caused by differences in expression level, organelle movement, photobleaching and variations in excitation intensity (Linders et al. 2022). To alleviate such errors, ratiometric pHLuorin (RpHLuorin) was developed, in which the ratio of emission intensities obtained at two excitation wavelengths normalises the signal for sensor concentrations (Miesenböck et al. 1998; Martinez et al. 2012). RpHLuorin2, a later variant with improved brightness, is the sensor used here (Mahon 2011).

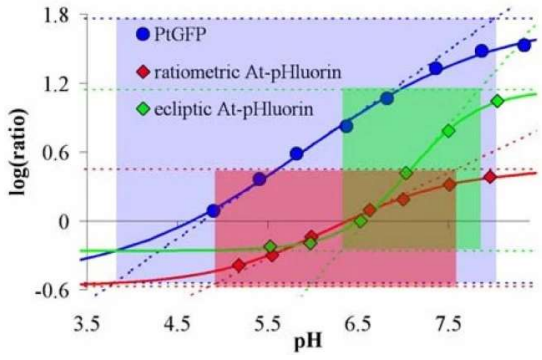


Figure 2. pH response (log ratio against pH) of PtGFP, ratiometric At-pHLuorin and ecliptic At-pHLourin. The coloured rectangles show the pH range of best responsiveness for each probe. Adapted from Schulte et al. (2006).

While the ratiometric method corrects for sensor concentration, it requires two sequential image acquisitions at different excitation wavelengths. This makes it slow, increases the chance of misalignment errors in highly dynamic organelles and leaves the ratio highly sensitive to background fluorescence, which can cause large variations in ratio values (Linders et al. 2022). Moreover, the fluorescence of GFP-derived proteins is reduced in acidic environments (Liu et al. 2006), so that less signal is collected, the signal-to-noise ratio (SNR) falls, and pH values in dim, acidic compartments can be inaccurate. Figure 2 illustrates the differences in pH response between the bio-sensing technologies.

1.3 Using FLIM for Quantitative Analysis

Fluorescence Lifetime Imaging Microscopy (FLIM) is a microscopy technique capable of generating a highly precise spatial map of fluorescence lifetime at the level of individual pixels (Torrado et al. 2024). When a fluorophore absorbs light, an electron is promoted to a higher energy level. When it subsequently returns to the ground state, a photon is emitted and fluorescence occurs. The fluorescence lifetime (τ) is the average time a fluorophore spends in the excited state before photon emission (Datta et al. 2020; Becker 2012). FILM with the ratiometric sensor RpHLuorin2 is capable of overcoming the limitations of previous biosensors by reading out pH from a pH-dependent shift in lifetime (Linders et al. 2022).

A critical advantage of FLIM is that τ is an intrinsic molecular property (Berezin and Achilefu 2010). To a first approximation, therefore, changes in fluorophore concentration, excitation light intensity and photobleaching do not affect its value (Jain et al. 2009), whereas τ remains sensitive to the local microenvironment, including pH, ion concentration, and molecular interactions. Furthermore, FLIM captures quantitative pH measurements in a single excitation scan using the pH-dependent lifetime shift of RpHLuorin2 (Linders et al. 2022) and is therefore also effective for capturing rapid cellular dynamics without errors or distortions due to organelle motion (Bower et al. 2018).

At 488 nm excitation, the fast-decaying neutral (phenol) form of the chromophore is excited mainly, with some contribution from the slower anionic (phenolate) form, so the observed decay is a mixture of the decays of two protonation states and the average lifetime rises with pH (Cotlet et al. 2001; Heikal et al. 2001; Linder et al. 2022). The effectiveness of FLIM with

RpHLuorin2 was demonstrated by Linders et al. (2022), who mapped the pH of the secretory pathway and found that the lifetime responded over a pH range of approximately 4.5 to 7.5, wider than the 5.5 to 7.5 range of ratiometric measurements (Figure 3). The sensor has been used to monitor drug-induced changes in cellular pH (Guo et al. 2025).

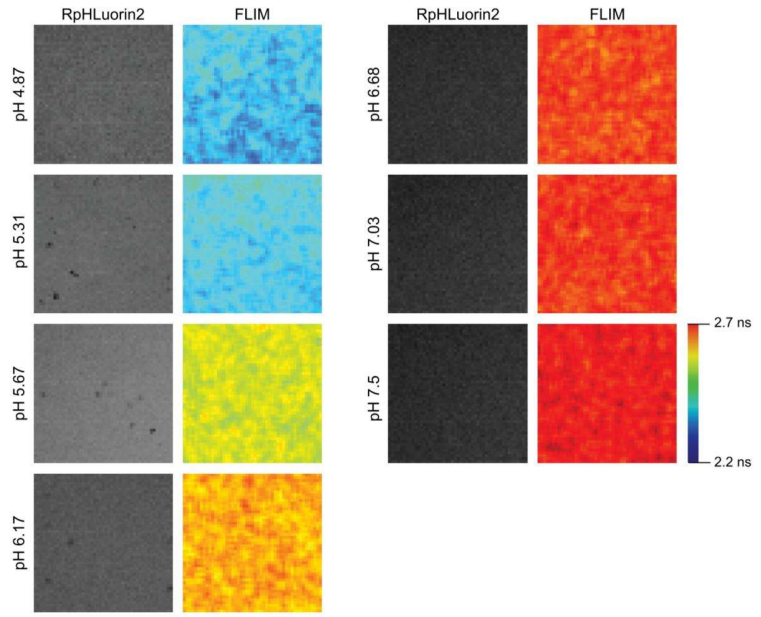


Figure 3. FLIM of recombinant RpHLuorin2 in calibration buffers of defined pH. Intensity images (left) and pseudo-coloured fluorescence lifetime images (right; colour scale 2.2 – 2.7 ns). Reproduced from Linders et al. (2022)

1.4 Data Analysis of the pH Heterogeneity of the Secretory Pathway

FILM datasets share their structure with hyperspectral images, in which a full spectrum is recorded at every pixel to form a three-dimensional data cube (Figure 4). In FILM, the time channels of the photon arrival histogram take the place of the spectral bands. Hyperspectral imaging has gained popularity due to its effectiveness when modified for analysis and the growing interest in electronic imaging (Khan et al. 2018). This has been particularly evident in pharmaceutical research, for applications such as spectroscopy and chemical imaging (Gowen and Amigo 2013), in food science, for chemometrics to analyse food structure, composition and quality (Amigo et al. 2013), and in biochemistry and biomedicine, for studies of biological samples (Zhang et al. 2014). The HIA (Hyperspectral Image Analysis)

program was developed to analyse hyperspectral images, especially for quantitative chemical imaging and unsupervised analysis of coherent anti-Strokes Raman scattering (CARS) data (Masia et al. 2013).

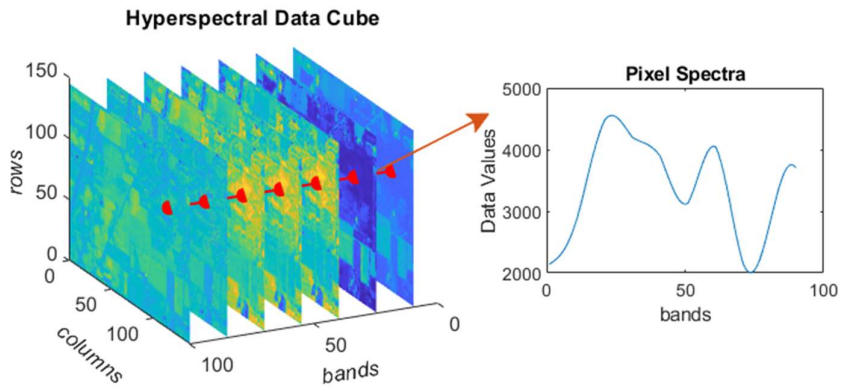


Figure 4. A hyperspectral data cube (rows x columns x bands). The spectrum recorded at a single pixel is extracted as a pixel-spectrum vector. In FILM, the bands correspond to the time channels of the photon arrival histogram. Adapted from MathWorks (n.d.).

While FLIM has been successfully used to determine the pH of the secretory pathway, it has not yet been optimally exploited for data analysis in a cellular context, as the fluorescence decay at each pixel does not follow a single rate constant. The data collected represent a linear combination of multiple components, including the pH sensor in various protonation states, cellular fluorophores (such as NADH and FAD) (Datta et al. 2020), and other co-expressed fluorophores. If the decay curves are fitted with standard mono-exponential or bi-exponential models, which require iterative and computationally demanding fitting, the model can impose an inappropriate functional form and perform poorly in regions of low photon counts (Warren et al. 2013; Datta et al. 2020). Linders et al. (2022) fitted a single exponential and reported an apparent average lifetime because of the limited number of photons.

To overcome such shortcomings in data analysis, several advanced computational methods have been developed. Phasor analysis uses a fit-free, graphical approach that transforms the lifetime decay at each pixel into a single point on a two-dimensional phasor plot (Digman et al. 2008; Ranjit et al. 2018). Although this method can quickly identify distinct components,

quantitative resolution becomes difficult when more than two components contribute or when lifetime are closely spaced unless additional constraints are applied (Datta et al. 2020). Another method is supervised unmixing or pattern matching, which decomposes the data into a linear combination of pre-defined reference decay patterns obtained from control samples (Gregor and Patting 2014). While this method can provide accurate analysis, it may lead to inaccuracies if the reference dynamics do not match those of the sample being analysed, owing to environmental differences between *in vitro* and *in cellulo* conditions.

The most advanced method, unsupervised unmixing FLIM (uFLIM), employs the Non-negative Matrix Factorisation (NMF) algorithm (Lee and Seung 1999) to separate a FLIM dataset into its constituent spatial distributions and temporal dynamics without any prior knowledge of the component decay profiles (Masia et al. 2022). NMF is a dimensionality reduction technique that factorises the data matrix D (pixels x time channels) into the product of a spatial concentration matrix S (pixels x components) and a temporal dynamics matrix T (components x time channels), as illustrated in Figure 5.

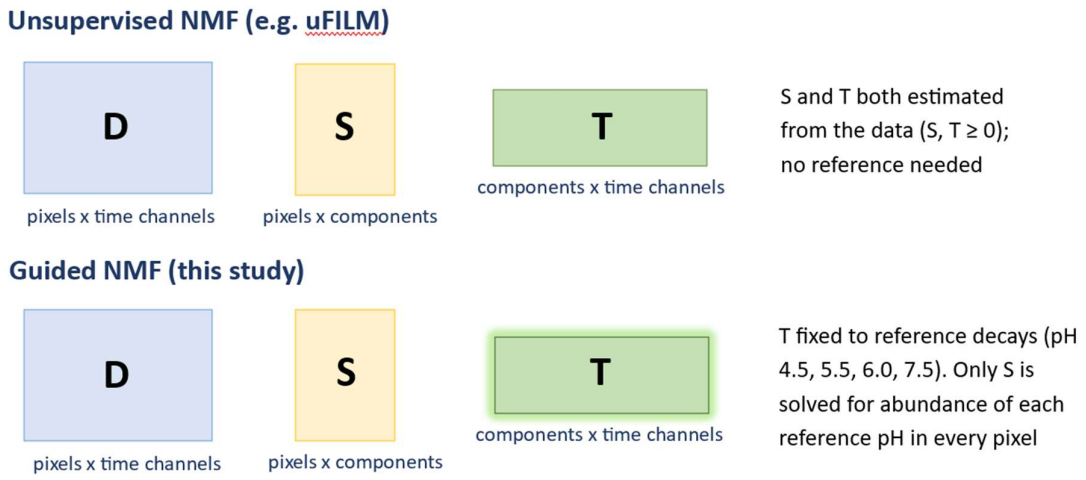


Figure 5. Schematic of NMF applied to FILM data. The data matrix D (pixels x time channels) is factorised into a spatial concentration matrix S (pixels x components) and a temporal dynamics matrix T (components x time channels). In unsupervised NMF, both S and T are estimated from the data. In guided NMF used here, T is fixed to reference decay curves and only S is solved for.

The non-negative constraint ensures that all results are physically interpretable, as the values in both matrices are greater than or equal to zero. uFLIM can be used to visualise different fluorescent components separately, as well as to reveal interaction parameters such as FRET rates, offering a powerful method for analysing complex biological samples. Unsupervised unmixing has also been used to quantify the distribution of probes between subcellular locations (Coelho et al. 2010), and model-free decompositions have been applied to protein transport in living cells (Wüstner 2022). A limitation of unsupervised factorisation is that NMF algorithms converge to local solutions that depend on initialisation, which becomes particularly important when the data are noisy (Boutsidis and Gallopoulos 2008).

1.5 Project Aims and Hypothesis

The main aim of this data analysis project was to propose a method for bridging the gap between currently available sensor technology and the limitations of computational analysis. A guided unmixing FLIM analysis was implemented to evaluate intracellular pH mapping. We hypothesised that the decay curves obtained from *in vitro* RpHLuorin2 could be used as effective references for guided analysis of cellular probe FLIM data, enabling the generation of accurate pH maps. The wider motivation was to move beyond simple lifetime fitting towards resolving heterogeneous pH microenvironments within organelles, and the pH changes experienced by trafficking cargo. This work establishes and critically tests the first, supervised, step towards that goal.

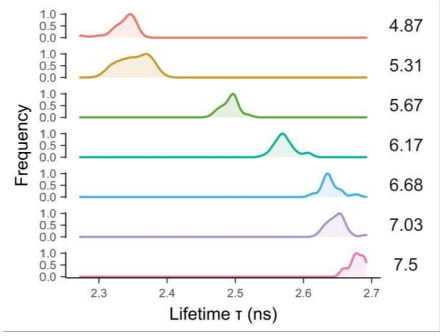


Figure 6. Average lifetime histograms of recombinant RpHLuorin2 at the calibration pH values, used to identify the key pH values for the reference set. Reproduced from Linders et al. (2022). The broad, overlapping distributions at some pH values indicate the limited SNR of per-region fitting.

To obtain the references required for guided analysis, the pH-dependent in vitro fluorescence lifetime decay of recombinant RpHLuorin2 was calibrated at specific key pH values that exhibited good peak separation (4.5, 5.5, 6.0, and 7.5), as illustrated Figure 6. These peaks correspond to distinct pH-dependent lifetime states (Linders et al. 2022). Selecting only pH values with good peak separation limits crosstalk between components. Reference decays with overlapping peaks are nearly collinear, so the algorithm cannot reliably apportion signal between them, which leads to inaccurate analysis.

Using these calibrated decays as fixed temporal profiles, they were implemented in a guided NMF algorithm to analyse FLIM data from cells expressing organelle-targeted probes. In guided NMF, the temporal dynamics matrix T is fixed to the pre-generated reference decay curves and the algorithm solves only for the spatial concentration matrix S. This determines how much of each reference pH component contributes to every pixel in the image.

After assessing the accuracy of the calculated pH against expected values, alongside the spatial resolution and sensitivity of the FLIM images, spatial pH maps were generated. By comparing our findings with the wider existing literature, we critically analyse the methodological limitations. This facilitates the proposal and justification of a new, integrated framework for quantitative pH mapping that combines the strengths of this project with future innovations to achieve clearer pH readouts.

2. **Experimental Procedures**

An overview of the analysis pipeline is shown in Figure 7. Recombinant calibration data were used to generate reference decays, which served as the fixed temporal matrix in guided NMF of the cellular FILM data. The resulting abundance maps were converted into pH maps and evaluated.

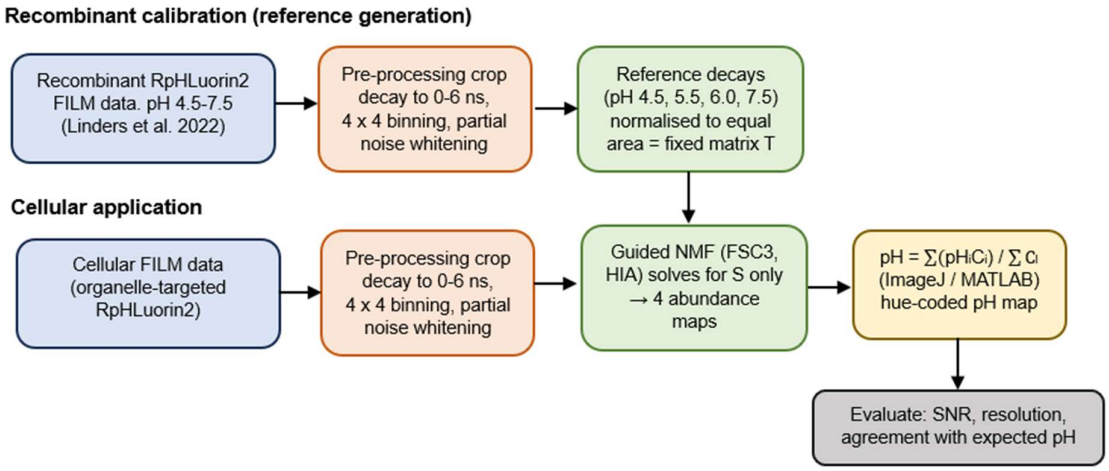


Figure 7. Overview of the analysis pipeline. Reference decays generated from recombinant RpHLuotin2 calibration data (top) define the fixed temporal matrix T used in guided NMF of cellular FILM data (bottom). Abundance maps are converted to pH maps by an abundance-weighted average and evaluated for SNR, resolution and agreement with the expected pH

2.1 Description of the FLIM Dataset

The recombinant and cellular FILM data analysed in this work were acquired by Linders et al. (2022), who deposited their raw data in the Zenodo repository. The acquisition procedure is summarised here for completeness. Recombinant RpHLuorin2 was expressed in *E. coli* and purified using nickel-nitrilotriacetic acid (Ni-NTA) affinity chromatography. The purified protein was diluted into calibration buffers of defined pH (nominally 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, and 7.5). The buffers contained 125 mM KCl, 25 mM NaCl and 25 mM of MES, or pH ≤ 6.5, or HEPES, for pH ≥ 7.0, and were equilibrated to the required temperature, with the pH of the final buffer (7.5) adjusted to 37°C. Cellular data were acquired from HeLa cells expressing RpHLuorin2 fused to organelle-specific targeting sequences or proteins (Linders

et al. 2022). The buffer composition itself can influence protein behaviour beyond pH (Salis and Monduzzi 2016).

FLIM data were acquired from each sample using a time-correlated single-photon counting (TCSPC) system equipped with a pulsed white light laser at an 80 MHz repetition rate and a HyD detector; a Leica SP8 SMD was used. The sensor was excited at 488 nm and emission was collected between 502 and 530 nm (Linders et al. 2022). For each pH buffer, a global fluorescence decay curve was generated by averaging the photons from the entire field of view. This creates a single, high SNR decay trace that accurately reflects the pH under the specified buffer conditions.

It should be noted that Linders et al. (2022) did not provide pixel size information. As a result, it was not possible to display scale bars with physical dimensions in the figures presented in this analysis, and binning factors are expressed in pixels rather than physical units. The absence of this information does not affect the analysis performed, as the methods employed operate on relative intensities and decay kinetics independent of absolute spatial scaling.

2.2 Data Pre-Processing and Optimisation

Before the analysis, several pre-processing steps were implemented, and their settings tested and optimised, to ensure data quality and to optimise the performance of the factorisation algorithm (The HIA software refers to the time axis of the decay as the spectral dimension, and this term is retained where it refers to software settings). Initially, examination of the full decay curves revealed an undesirable secondary peak at approximately 7-8 ns after the initial excitation peak (Figure 8). This peak was judged to arise from an instrumental artefact rather than genuine pH-dependent fluorescence decay. To ensure that only the relevant sensor decay was analysed, the range was limited to 6 ns for all subsequent analyses. This cropping removes the artefact while preserving the main part of the decay (0 to 6 ns), which carries the pH information, at the cost of discarding the remainder of the 12.5 ns repetition period.

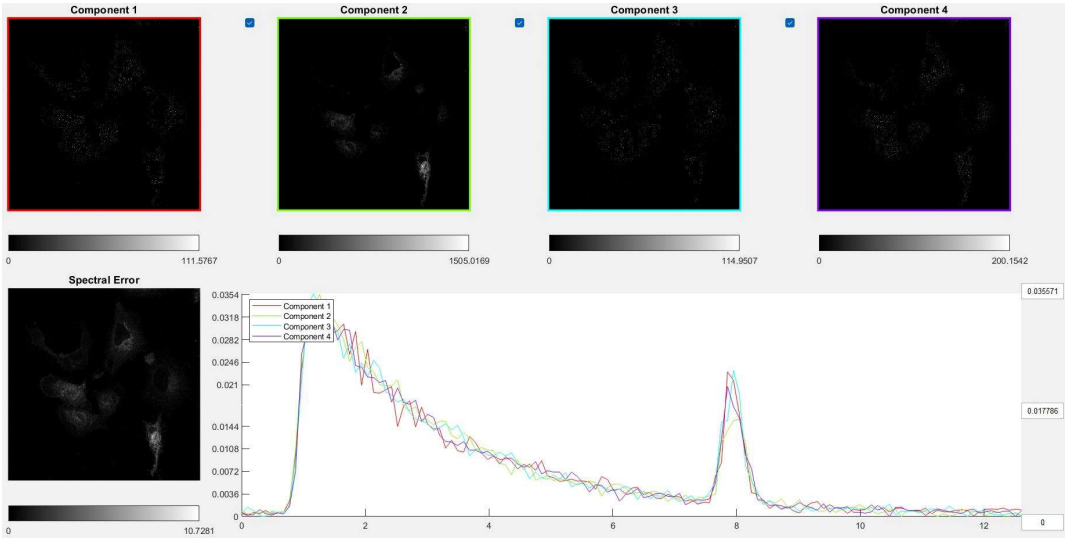


Figure 8. Analysis result of the pH 4.5 probe dataset without limiting the spectral range and without binning. A second peak is visible at approximately 7-8 ns, after the main decay.

Signal-to-noise ratio (SNR) was optimised through spatial binning. SNR is a metric that compares the power of the signal to the background noise: a higher SNR means that even small fluctuations in pH can be detected, while a lower ratio means that the data are obscured by noise. Spatial binning mitigates low photon counts by summing adjacent pixels, which increases the number of photons per analysed unit and so reduces the relative shot noise. Various binning sizes, ranging from 1 x 1 to 128 x 128, were tested. A binning of 4 x 4 was found to be optimal, as it provided a significant increase in SNR per image element and the most accurate pH reading without excessive loss of spatial resolution. The effect of different binning factors on image quality is presented in Figure 9.

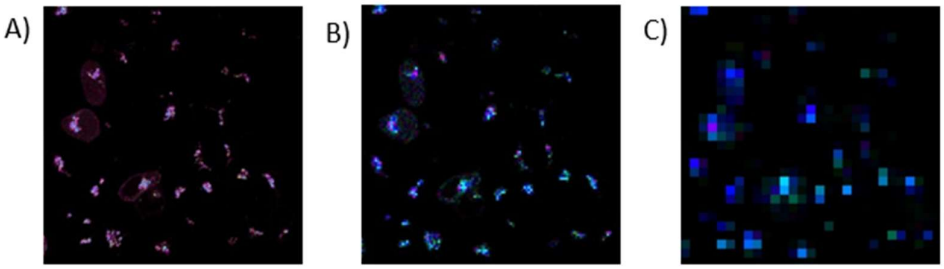


Figure 9. Colour-coded pH maps of pH6.5 probe dataset obtained with different binning sizes. A) 1 x 1 binning gives poor SNR.
B) 4 x 4 binning gives good SNR without loss of resolution.
C) 16 x 16 binning begins to lose spatial resolution.

Partial noise whitening was applied to improve the stability of the factorisation. FILM data are acquired by photon counting, so the noise follows Poisson statistics: bright signals, found at the initial peak of the decay curve, have a higher absolute noise variance than dim signals at the tail of the decay. Noise is therefore not uniform across the dataset, and the bright early channels contribute disproportionally to the factorisation error, even if their relative error is small. Noise whitening, a pre-processing step that is standard for Poisson-distributed spectral data (Keenan and Kotula 2004), rescales the data to a more uniform variance before factorisation. This prevents the algorithm from being dominated by the high variance in the early part of the decay curve and, more importantly, allows it to analyse the low-signal region of the decay, which encodes subtle lifetime differences. The effect of partial noise whitening on data quality is illustrated in Figure 10.

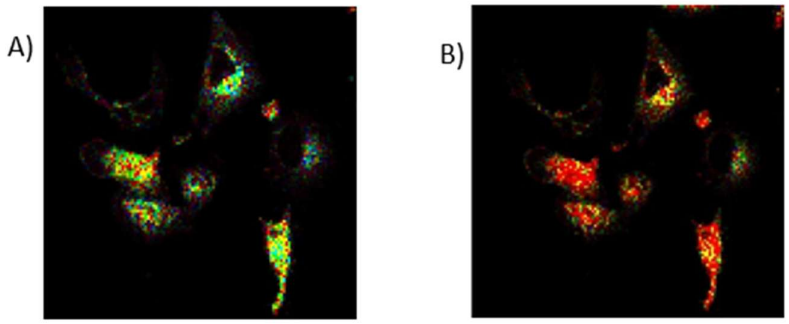


Figure 10. Analysis results with and without partial noise whitening. A) Partial noise whitening applied B) Partial noise whitening not

2.3 Generation of Reference Data for Guided Analysis

After the pre-processing step, reference decay curves were generated from the recombinant calibration data to serve as fixed components for guided non-negative matrix factorisation (NMF) analysis.

The four pH values (4.5, 5.5, 6.0, and 7.5) exhibiting the greatest spectral variation were selected from the completed calibration set and imported into the Hyperspectral Image Analysis (HIA) software (Masia et al. 2013; Masia et al. 2022). The remaining calibration samples were not used as references, which allowed the pH 5.0 and 6.5 samples to serve as independent test samples (Section 3.3). For each of the four reference datasets, the spectral average decay curve was extracted and normalised to have equal integrated area. This

normalisation ensures that each reference curve represents a probability density function for photon emission at that pH, independent of absolute intensity variations.

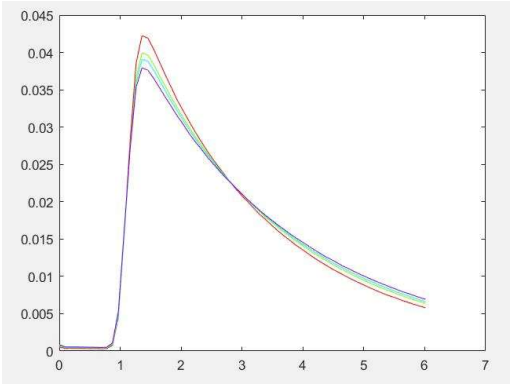


Figure 11. Normalised reference decay curves (pH 4.5, 5.5, 6.0, and 7.5) used as the fixed temporal matrix T for guided analysis. Curves are normalised to equal integrated area.

In a new workbook, the calculated normalised curve data were compiled to obtain four probability density functions representing the normalised decay fingerprint of each reference pH (Figure 11). These four curves were assembled into a single ASCII text file to act as the fixed reference matrix for all subsequent guided analyses. Because the reference and the samples were measured with the same instrument, the references implicitly contain the instrument response function, which therefore does not need to be deconvolved separately (Gregor and Patting 2014).

2.4 Non-Negative Matrix Factorisation (NMF) Framework

NMF is the algorithm at the core of the analytical approach used in this study. It works by reducing the dimensionality of the data and separating the source image into additive components. The fundamental principle of NMF can be expressed as $D \approx S \times T$, where D is the original data matrix (pixels x time channels), S is the spatial concentration matrix (pixels x components), and T is the temporal dynamics matrix (components x time channels) (Lee and Seung 1999) (Figure 5).

For FLIM data analysis, the decay curve measured at each pixel can be interpreted as a linear combination of a small number of component decay curves. Each component has a specific decay profile (stored in T) and a spatial distribution indicating how much of that component is present in each pixel (stored in S).

A non-negativity constraint is important for good physical interpretability. Unlike other factorisation methods, such as principal component analysis, which can yield negative values, NMF forces all values in both S and T to be greater than or equal to zero. This accords with the fact that fluorescence measurements count photons, which cannot be negative, so the contribution of a component to a pixel's signal cannot be negative. The constraint ensures that the decomposition gives interpretable results (Lee and Seung 1999).

In our analysis, a guided NMF approach was used. Instead of an unsupervised approach, in which allows the algorithm calculates both S and T from the data, the T matrix was fixed using the pre-generated reference decay curves corresponding to pH 4.5, 5.5, 6.0, and 7.5. The algorithm was used only to determine the spatial concentration matrix S, which gives the contribution of each reference pH component to each pixel in the image. With T fixed, estimating S at each pixel reduces to a non-negative linear regression problem, which has a unique solution when the reference decays are linearly independent. This makes the analysis stable, but ties its results to the chosen references.

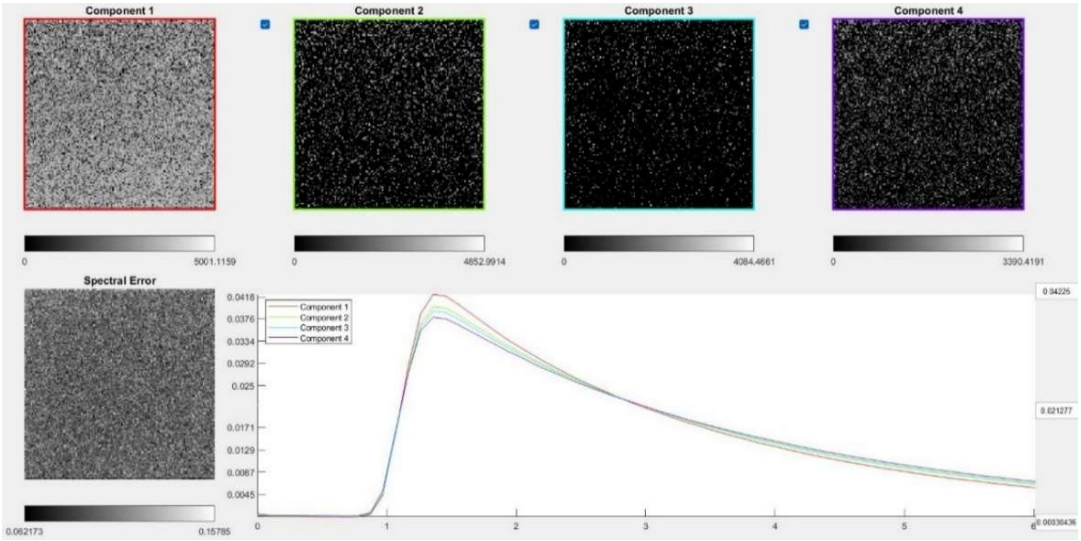
2.5 Guided NMF Analysis of Calibration and Cellular Data

Using the determined framework, guided NMF analysis was performed on both the recombinant and probe datasets. In the HIA software, a FLIM dataset of recombinant protein with a known pH value was selected and loaded. The spectral range was limited to 6 ns, and the image was binned with the x- and y-axes set to 4. The data were then processed using FSC3 (Factorisation into Susceptibilities and Concentrations of Chemical Components) with the number of components set to 4, and the minimum and maximum iterations set to 1. FSC3 is a core component of the HIA software, developed to analyse hyperspectral Coherent anti-Stokes Raman Scattering (CARS) data and to generate concentration maps of chemical components using NMF (Masia et al. 2013; Boorman 2019). The number of iterations is irrelevant here, as the software performs a projection that determines only the

abundance maps of the components defined by the supplied decay curves. The single method was selected, as the knock-out method is typically used for calculations in which no guide is provided. Partial noise whitening and guided analysis were selected and, when prompted, the ASCII text file containing the four reference decay curves was chosen.

The output from FSC3 processing was a stack of four images, with the reference decay graphs, corresponding to the spatial concentration maps for each reference pH component. By changing the normalisation setting from Concentration to Spectrum Integral, the dominant component and the differences between components were observed and noted. Figure 12 shows the results of analysing the pH 5.0 and 6.5 recombinant datasets. Each output was saved.

A)



B)

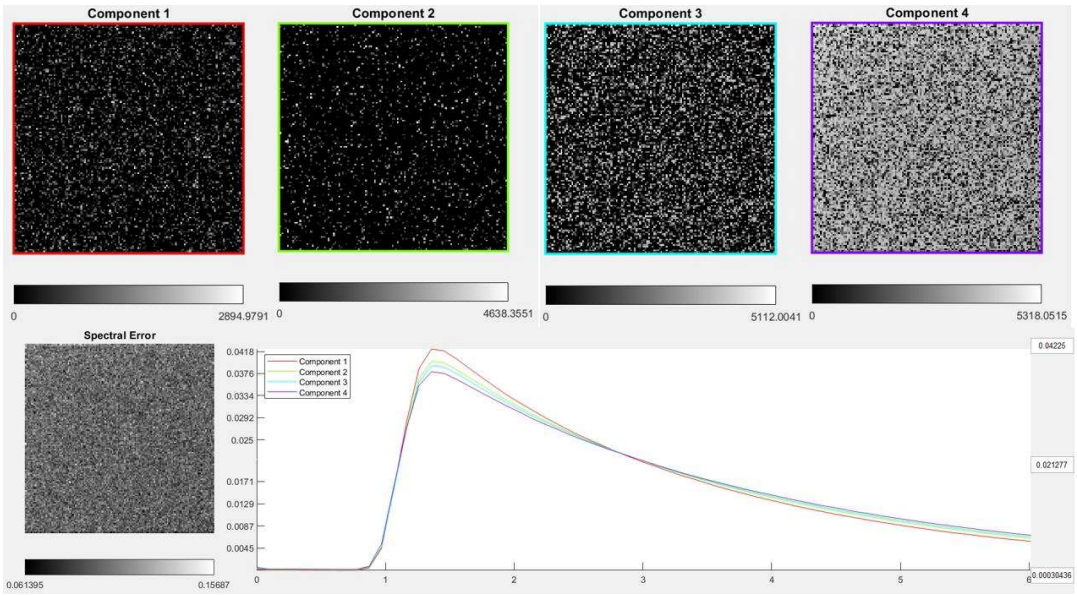


Figure 12. Image matrix showing the four components maps (spatial concentration maps). A) pH 5.0 recombinant dataset, with component 1 dominant. B) pH 6.5 recombinant dataset, with component 4 dominant.

The same guided NMF procedure was then applied to the cellular probe datasets. Probe data are FILM data acquired from living cells expressing organelle-targeted RpHLuorin2 constructs, used to map the pH inside organelles. The spectral range was limited to 6 ns and binning was set to 4 x 4. FSC3 was run with 4 components, minimum and maximum iterations of 1, the single method, guided analysis and partial noise whitening, and with the ASCII reference file supplied. The result for a pH 4.5 probe dataset is shown in Figure 13, which presents four images corresponding to the spatial concentration maps of the four reference pH values.

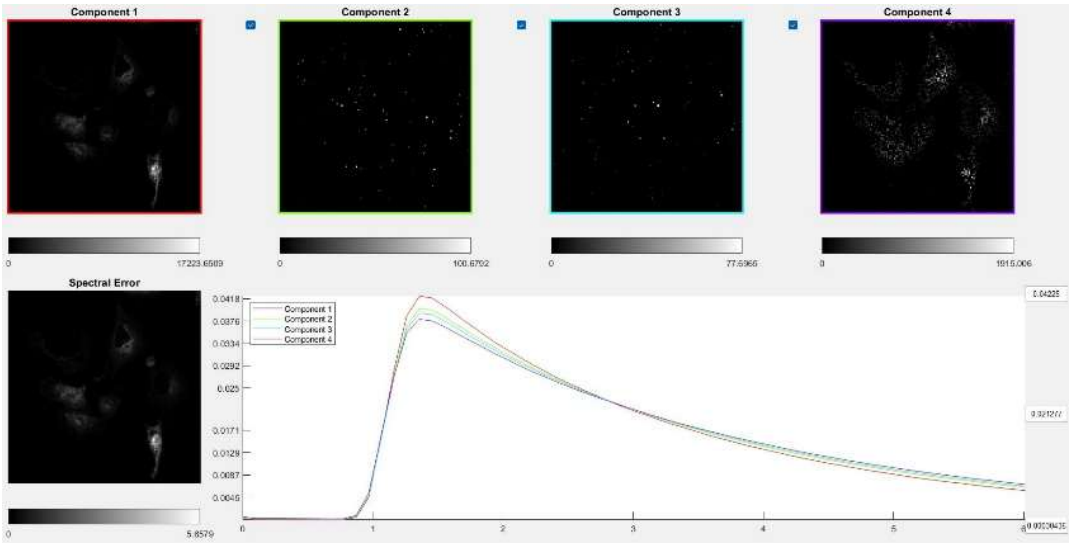


Figure 13. Image matrix showing the FSC3 results of a pH 4.5 probe dataset. Component 1 is dominant.

2.6 Calculation and Visualisation of Quantitative pH Maps

The concentration maps generated by NMF were processed further to generate quantitative pH maps using a weighted average approach. The saved concentration maps, which indicate how much each pH component contributes to each pixel (Figure 12), were imported into

ImageJ for quantitative synthesis (Schneider et al. 2012). For each pixel, a weighted-average pH was calculated using the formula:

$$pH_{estimated} = (\sum_i pH_i C_i)/(\sum_i C_i)$$

(where i is the component index, pH_i is the nominal pH of each reference component (4.5, 5.5, 6.0, 7.5), and C_i is the abundance (concentration) of that component in the pixel). Because the references are normalised to equal area, the C_i are fractions of detected photons rather than fractions of sensor molecules, and the estimate is bounded by the lowest and highest reference pH by construction.

To make visualisation of the pH values easier, a custom MATLAB script was employed. The script calculated the pH using the weighted average formula and mapped the resulting pH value to a hue value: the pH range of 4.5 – 7.5 was linearly mapped to a hue range of 0 - 1. The normalised sum of the concentration maps was used to determine the value (brightness), with saturation kept at 1 for all pixels. The resulting HSV (hue, saturation, value) image was converted to RBG for display (Figure 14). This colour-coded representation allows pH distribution across the sample to be assessed visually. Rainbow-type hue maps do not vary uniformly in perceived lightness and can be difficult to read for people with colour-vision deficiency (Crameri et al. 2020), so the colour bar is always shown alongside the maps and perceptually uniform colour maps are recommended for future work.

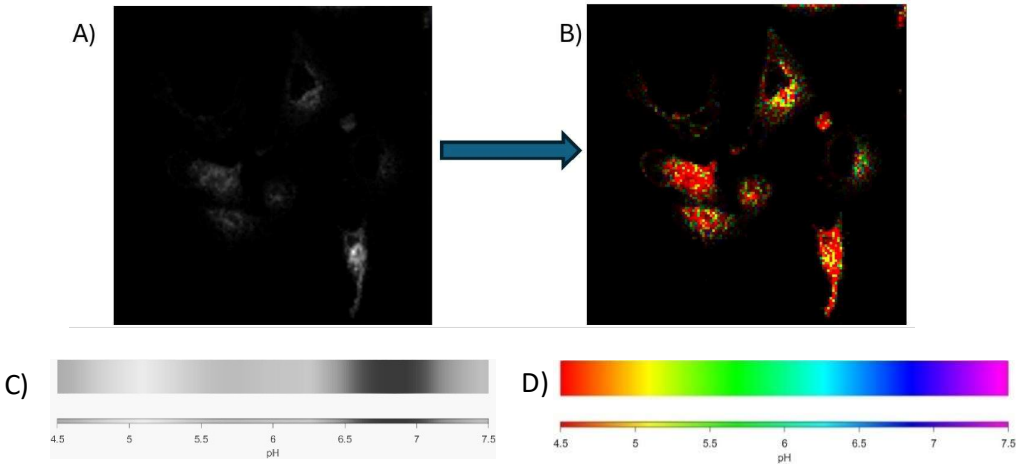


Figure 14. Visualisation of the pH map of a pH4.5 probe dataset. A) Grayscale (intensity) image. B) Colour-coded pH map, showing predominantly red hues. C) Grayscale bar D) pH colour bar

3. **Results**

3.1 Impact of Partial Noise Whitening on Data Quality

The effect of partial noise whitening on the quality of the factorisation was assessed by comparing results obtained with and without this pre-processing step. As illustrated in Figure 10, analysis performed without partial noise whitening resulted in pH maps containing excessive noise and poor spatial consistency. This is consistent with the algorithm being dominated by the high variance in the early decay region, which prevented it from extracting meaningful component distributions from the data (Keenan and Kotula 2004). When partial noise whitening was applied, the resulting pH maps showed improved clarity, with more coherent spatial structures and reduced pixel fluctuations. This confirms the importance of appropriate noise handling for obtaining stable NMF solutions from photon-counting FLIM data. The comparison is qualitative and was made on a single dataset: a quantitative matric, such as the variance of the estimated pH within a region of uniform pH, would be needed to establish the size of the improvement.

3.2 Effect of Spatial Binning on pH Map Quality

The importance of spatial binning or obtaining a good quality pH map was studied. Figures 9 and 15 show comparisons of results obtained with different binning factors. Analysis performed without binning (1 x 1) produced pH maps dominated by noise, as seen in Figure 15(A) and (B), with large variations between pixels that rendered the data unreliable for determining pH values. As the binning factor was increased, image quality and signal-to-noise ratio (SNR) progressively improved. At 4 x 4 binning, the pH map showed clear spatial coherence with regions of uniform pH (Figure 15(C)). At binning factors of 8 x 8 and above, while the SNR continued to improve, spatial resolution was lost, with fine structural details becoming increasingly blurred. Based on this evaluation, 4 x 4 binning was selected as the best compromise between SNR improvement and preservation of spatial resolution, was used for all subsequent analyses.

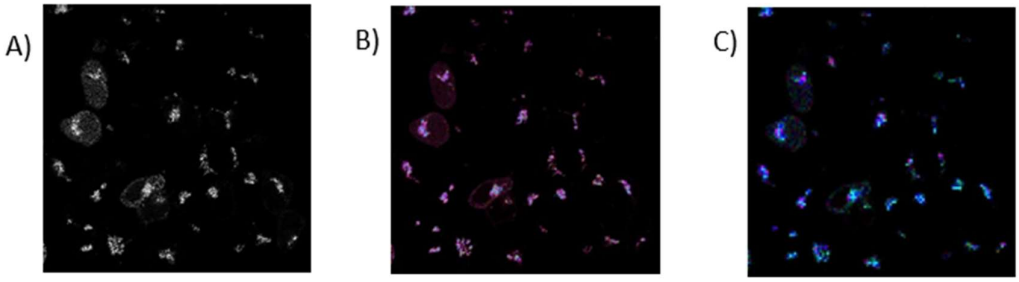


Figure 15. A) Greyscale result without binning. B) Colour-coded result without binning. C) Result with 4x4 binning, which gives a better SNR.

3.3 Validation of Guided Analysis Using Recombinant Calibrated Data

To ensure that the analysis framework performs as expected, without systematic errors, the guided NMF method was first tested using recombinant RpHLuorin2 datasets of known pH. The pH 4.5 and 7.5 datasets also contributed to the reference set, so agreement for these two samples is a self-consistency check. The pH 5.0 and 6.5 samples were not used as references and therefore provide an independent test of interpolation between references. When the pH 7.5 recombinant sample was analysed, the resulting pH map was in line with the expected outcome. After RGB colour mapping the image was violet, the hue assigned to high pH in this colour scheme (Figure 16(D)), and histogram analysis of this result showed a peak at approximately pH 7.3. The assignment of colours to pH values depends on the specific look-up table (LUT) employed. For this analysis, violet represented higher pH and red represented lower pH, and this mapping was consistently applied throughout all analyses.

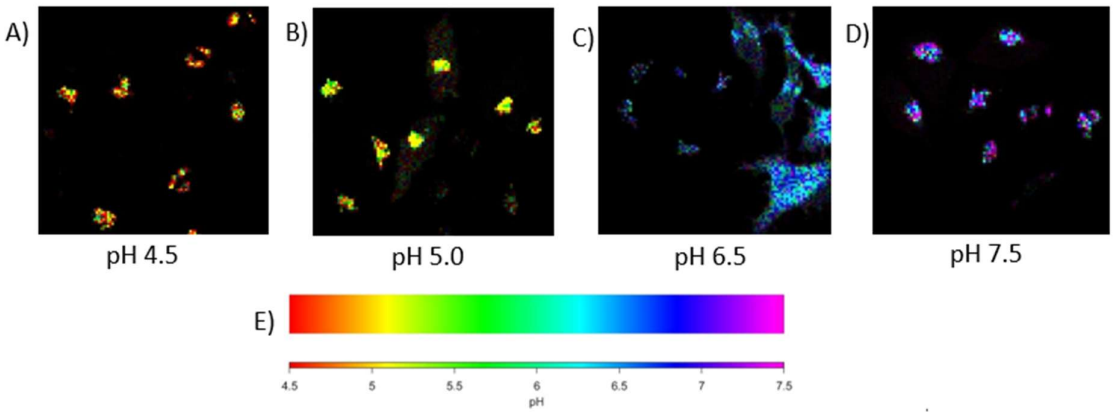


Figure 16. Colour-coded pH maps of recombinant samples, showing that the algorithm assigns pH on the basis of the reference decays. A) Predominantly red, pH 4.5. B) Predominantly yellow, pH 5.0. C) Predominantly light blue, pH 6.5. D) Predominantly violet, pH 7.5. E) pH colour bar.

When the pH 4.5 recombinant sample was analysed, the resulting pH map was predominantly red (Figure 16 (A)), the hue assigned to low pH, and the histogram peaked at pH 4.7. This demonstrates that the algorithm assigns pH values correctly on the basis of the reference decay curves. Furthermore, the intermediate pH recombinant samples (5.0 and 6.5) were also analysed, each showing colour distributions consistent with its expected pH, as illustrated in Figures 16(B) and (C). In these samples, the dominant component was component 1 for pH 5.0 and component 4 for pH 6.5 (Figure 12), that is, the nearest available reference in each case (assuming the components are numbered in the order of the reference file). The offsets observed at the two ends of the range (+0.2 at the pH 4.5 and -0.2 at pH 7.5; Table 1) are both directed towards the centre of the calibrated range. This is expected for an abundance-weighted average, which cannot lie outside the range of the reference pH values, so that any noise that places abundance in the wrong component pulls in the estimate inwards.

Table 1. Agreement between nominal pH and the peak of the pH histogram for recombinant RpHluorin2 datasets analysed by guided NMF. Offset = histogram peak minus nominal pH. Samples at pH 4.5 and 7.5 were also used to generate the references (self-consistency check); pH 5.0 and 6.5 were held out.

Nominal pH	Role in Analysis	Histogram Peak (pH)	Offset (pH units)
4.5	Reference	4.7	+0.2
5.0	Held-out test	5.2	+0.2
6.5	Held-out test	6.4	-0.1
7.5	Reference	7.3	-0.2

3.4 Application to Cellular Probe Data

After successful validation on the calibration data, the guided analysis method was applied to probe FLIM data acquired from living cells expressing organelle targeted RpHLuorin2 constructs. Each cellular probe dataset has an expected pH value based on the targeted organelle (for example ER ~ 7.2, Golgi ~ 6.0 - 6.5). The MATLAB script was used to encode pH as colour, generating colour-coded pH maps for visual assessment of their values.

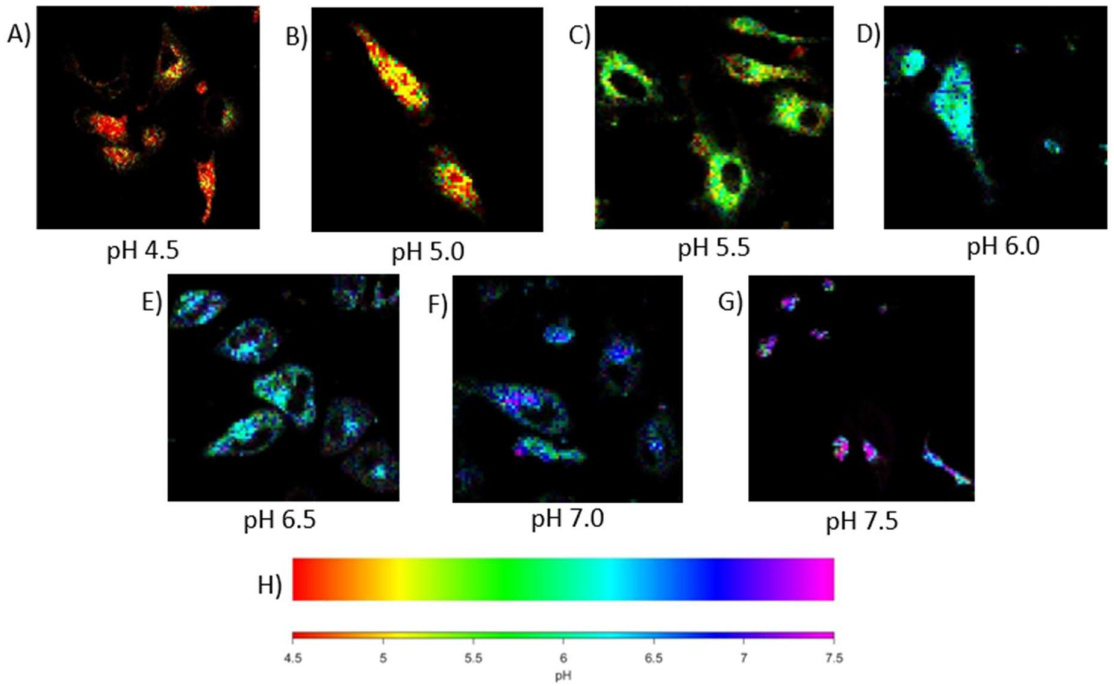


Figure 15. Colour-coded pH maps of selected cellular probe datasets that showed good agreement with the expected organelle pH. Colours correspond to pH according to the colour bar

Figure 17 presents a selection of cellular probe datasets that showed good agreement between calculated and expected pH values. The colour distributions across these images were consistent with the anticipated pH ranges for the targeted organelles, each showing coherent spatial structures. The sample targeted to the ER in figure 17(G) displays predominantly violet hues consistent with near-neutral pH, while those targeted to the Golgi in figure 17(D) and (E) show blue-green colours corresponding to a moderately acidic environment. The lysosomal sample in Figure 17(A) exhibits red hues consistent with highly acidic conditions. These results show that the guided NMF method can generate interpretable pH maps from cellular FLIM data when experimental conditions are favourable. Because Figure 17 shows datasets for good agreement, it should not be taken as representative of all datasets; [n] of [N] datasets fell into this category.

However, some cellular probe datasets produced results that deviated from the expected values, as seen in figure 18. In these cases, the colour-coded pH maps were inconsistent with the expected pH of the organelle and some images showed poor spatial structure with

increased fragmentation. It is important to note that the analysis procedure applied to these datasets was identical to that used for the successful examples in Figure 17, with the same binning factor (4 x 4), reference curves, and guided NMF parameters. The discrepancies therefore cannot be attributed to a difference in the analysis procedure.

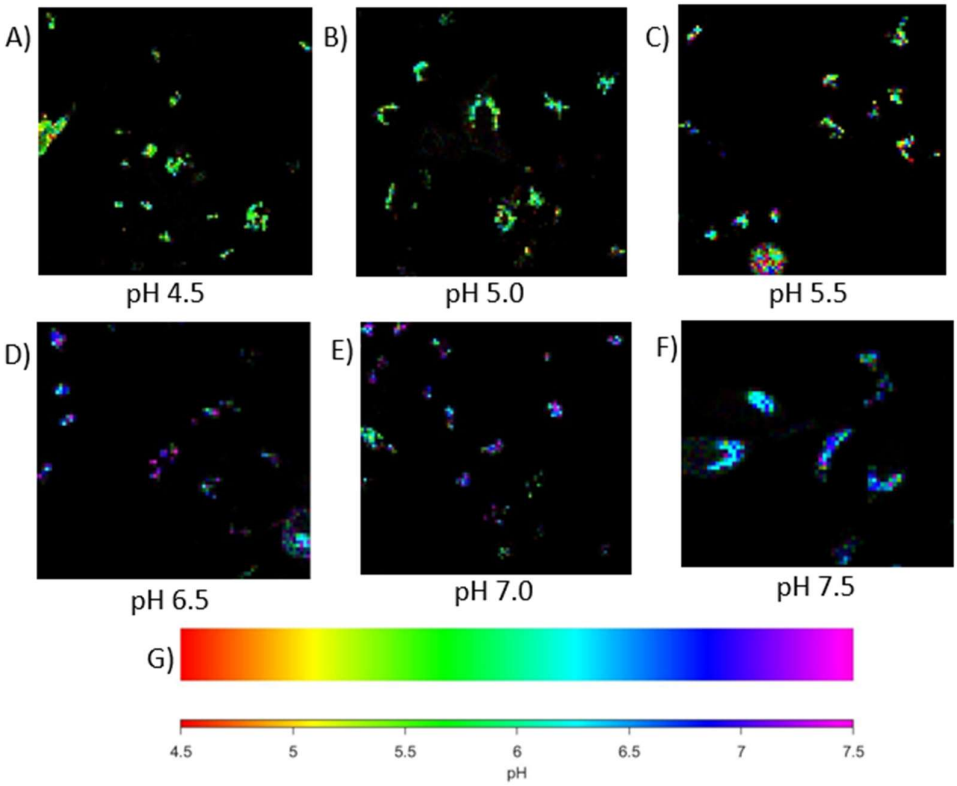


Figure 18. Cellular probe datasets for which the analysis gave poor spatial structure and colours inconsistent with the expected pH.

A possible explanation for this variability is that the datasets shown in Figure 17 and figure 18 were acquired on different experimental days, and day-to-day variation in experimental conditions could affect data quality. Other explanations include lower excitation power, resulting in reduced photon counts and poorer SNR, incomplete pH stabilisation of the cellular environment, or slight misalignment of the optical system, all of which could degrade data quality. Photon counts are also particularly low for ER-targeted probes, whose signal is diffusely distributed through the cell. Linders et al. (2022) observed that low photon counts in the ER led to an overestimation of pH. Without detailed information on the acquisition conditions of each dataset (photon counts per pixel, excitation power and acquisition date), it is not possible to identify the cause of the variability.

However, since some datasets produced excellent results while others did not, despite identical analysis procedures, the variable results are more likely to reflect raw data quality than the analysis procedure, although an interaction between the two cannot be excluded, as a fixed-reference model is more sensitive to low photon counts than a flexible one. This highlights the importance of consistent experimental conditions and sufficient photon budget for reliable FLIM pH mapping (Nasser and Meller 2021).

4. **Discussion**

4.1 Integration of Experimental Findings with Existing Literature

The results of this project both validate and critically qualify principles from the existing literature on FLIM data analysis. The method was successful in differentiating between a broad range of pH values and was particularly effective at classifying samples as acidic or neutral. Validation on recombinant calibration data showed that the algorithm assigns pH on the basis of the reference decay curves across the range 4.5 – 7.5, with histogram peaks within 0.2 pH units of the nominal values at both extremes. This supports the foundational principle that the lifetime of RpHLuorin2 is a reliable proxy for pH and that multi-component analysis can yield spatially resolved pH states, in agreement with Linders et al. (2022). Since two of the four validation samples were also reference samples, however, the more informative test is the behaviour at the held-out pH values, where the dominant component was the nearest reference, so that resolution of intermediate pH values within the reference spacing (0.5 to 1.5 pH units) rests on the weighted average, which is a heuristic (Section 4.4).

The successful generation of interpretable pH maps from cellular probe datasets (Figure 17) shows that the guided NMF approach can be applied to complex biological samples. The ER-targeted sample displayed violet hues consistent with near neutral pH, the Golgi-targeted samples showed intermediate colours corresponding to their moderately acidic environment, and the lysosomal sample exhibited red hues consistent with highly acidic conditions. These results support the practical use of the method for organelle pH mapping when experimental conditions are favourable.

However, the observation that some cellular datasets produced poor results despite identical procedures (Figure 18) means that caution must be exercised. The photophysical behaviour of a fluorophore *in cellulo* can differ from that observed *in vitro* because of macromolecular crowding, viscosity, and specific local interactions (Wu and Minton 2015). Linders et al. (2022) themselves found that the absolute lifetimes of RpHLuorin2 in cells were lower than those of the recombinant protein, which they attributed to crowding-related self-quenching or differences in the local microenvironment, and they therefore calibrated the probe in cells using a GPI-anchored construct. A supervised approach employs *in vitro* references as a guide is vulnerable to such divergence when the reference dynamics

imperfectly match cellular conditions. This directly aligns with the warning by Masia et al. (2022) regarding the risks of using fixed reference patterns when the environments of reference and experimental samples are not perfectly matched.

The variability between successful and unsuccessful cellular datasets cannot be attributed to the analysis procedure alone, as the same procedure was applied to every analysis. The datasets shown in figure 17 and figure 18 were acquired on different experimental days, suggesting that even minor variations in experimental conditions can affect data quality. Lower excitation power, incomplete pH stabilisation of the cellular environment, or slight optical misalignment could all reduce data quality, but the cause cannot be resolved without the acquisition metadata. This again highlights the importance of consistent experimental conditions and sufficient photon budget for reliable FLIM pH mapping.

Several limitations of this study should be stated explicitly. The datasets were acquired by others, so acquisition metadata such as photon counts per pixel, excitation power and acquisition date were not available. Validation was largely qualitative (colour maps and histogram peaks) and based on a single dataset per pH value. The cellular results in Figure 17 are a selection, and no independent in-cell ground truth was available for the analysed datasets. The conclusions are therefore those of a proof of principle rather than a calibrated measurement.

4.2 The Photon Budget and Its Implications

The necessity for spatial binning highlights one of the major constraints in FLIM, which is the photon budget. As demonstrated in figure 15, analysis performed without binning (1 x 1) produced pH maps dominated by noise, with large pixel-to-pixel that rendered the data unreliable for quantitative interpretation. Only with 4 x 4 binning did the pH maps show clear spatial structures with well-defined regions of uniform pH.

The precision of parameters extracted from photon-shot-noise-limited data scales with $1/\sqrt{N}$, where N is the photon count (Köllner and Wolfrum 1992; Gerritsen et al. 2002). Binning b x b pixels increases N by a factor of b^2, and therefore improves precision by a factor of b while degrading the linear spatial resolution by the same factor. 4 x 4 binning

thus gives a fourfold gain in precision for a fourfold loss in linear resolution. The requirement to bin data to obtain meaningful results therefore directly sacrifices spatial resolution to achieve a reliable lifetime reading, or SNR, per analysed unit (Callenberg et al. 2021). Such a trade-off must be carefully considered when designing live-cell FLIM experiments.

The same constraint applies to the discrimination of intermediate pH values. The lifetimes of RpHLuorin2 differ by only a few tenths of a nanosecond across the pH range (Linders et al. 2022), so the reference decays are very similar in shape, and many photons are needed to apportion signal reliably between them (Nasser and Meller 2021). The optimal binning factor of 4 x 4 represents the best compromise between SNR improvement and spatial resolution preservation for this dataset, but it should not be assumed to transfer to others.

Noise handling is an important part of data analysis, and this was further emphasised by the results obtained with partial noise whitening (Figure 10). Analysis performed without whitening resulted in pH maps containing excessive noise and poor spatial coherence, as the algorithm was dominated by the high variance in the early decay region, whereas whitening improved clarity. This confirms that testing for the best conditions and settings is essential for NMF solutions of photon-counting FLIM data, consistent with the approach used to validate the uFLIM algorithm (Masia et al. 2022).

4.3 Background Noise: An Ongoing Issue

Noise was consistently observed in the pH image maps as a factor affecting data quality. It arises where objects are close to each other, overlapping, or touching during spectral imaging (Amigo et al. 2015). Moreover, organelles exhibit different spectral profiles, which directly affect the spectra of the background adjacent to the object being imaged, potentially causing the two to share similar spectral characteristics. This makes the removal of noise by automated methods challenging, as differentiating between signal and background becomes difficult.

To mitigate the noise issue, an identification mark could be assigned to each pixel belonging to the structure under analysis, classifying them into the same class. All other pixels would

then be removed, leaving an image free of background noise. This method, proposed by Amigo et al. (2015), would however be extremely time consuming and highly dependent on correctly identifying the organelle and its boundaries, introducing the potential for user bias. Denoising approaches that exploit redundancy across pixels and time channels, such as principal component analysis of FILM data (Soltani et al. 2025), offer an alternative that does not require manual segmentation. Since our analysis is local, meaning that the result for a pixel does not affect the results of other pixels, we would only experience an issue in overlapping regions and not in pixels within the same organelle. This independence of pixel-wise analysis is both a strength, helping to avoid error propagation, and a limitation, due to the inability to use spatial context for noise reduction (Kervrann et al. 2015).

4.4 The Conceptual Framework of NMF and its Application

The NMF algorithm used in this analysis provides an interpretable framework for unmixing complex FLIM data. It enables the separation of the signals contributed by different sources to the decay curve at each pixel. In the guided NMF approach, the temporal dynamics matrix T was fixed using the pre-generated reference decay curves corresponding to pH 4.5, 5.5, 6.0, and 7.5. These pure-component decays act as a guid that improves the stability and interpretability of the decomposition, and the algorithm solves only for the spatial concentration matrix S which determines how much of each reference pH component contributes to every pixel in the image. The resulting concentration maps, as seen in Figures 12 and 13, help to visualise the spatial distribution of each pH state. Calculating the weighted average then converts the four component maps into a single quantitative pH map.

It should be noted that an initial analysis attempt was made using the unsupervised method, where no reference spectra were provided and both S and T were allowed to change freely. This approach did not give interpretable results, which is likely to reflect the low photon budget and the subtle differences between closely spaced pH states, since unsupervised NMF is sensitive to noise and initialisation (Boutsidis and Gallopoulos 2008). This led to the adoption of the guided approach, in which reference component spectra were used to constrain the solution. The unsupervised attempt was not evaluated quantitatively here.

The conversion of abundance maps into pH by a weighted average also deserves comment. The estimate is bounded by the lowest and highest reference pH, so it cannot reach 4.5 or 7.5 exactly and is pulled inwards by noise at the ends of the range, in agreement with the offsets of +0.2 and -0.2 observed in Table 1. The abundances are photon fractions rather than molecular fractions, and the lifetime-pH relationships of RpHLuorin2 is sigmoidal (Linders et al. 2022), so the mixing of two protonation states with different brightness does not translate linearly into pH.

A more principled alternative would be to fit the abundances with a two-state (protonated and deprotonated) model whose mixing fraction follows the Henderson-Hasselbalch relationship, which would recover pH from a single, physically meaningful parameter. Finally, the four-component model is over-parameterised. The reference decays differ only subtly in shape and are therefore strongly correlated, and the intermediate references (5.5 and 6.0) contributed minimally to the final result. A two-reference model using the pH extremes may therefore be sufficient and more robust.

4.5 The Pathway Forward: Moving from Supervised Guidance to Unsupervised Discovery

The limitations identified in this study suggest the development of a new analytical framework, representing a logical progression from our supervised, model-constrained approach to unsupervised, data driven approaches such as uFLIM (Masia et al. 2022) and phasor S-FLIM (Digman et al. 2008; Scipioni et al. 2021). Implementing the uFLIM algorithm on cellular FLIM data could address the rigidity of the analysis model. Unlike guided NMF, which fixes component dynamics to *in vitro* references, uFLIM determines both the spatial distribution and the temporal dynamics from the data being analysed, without requiring pre-defined reference spectra.

This unsupervised approach could also analyse the decay of organelle-targeted RpHLuorin2 without bias, potentially revealing intracellular environmental heterogeneity within an organelle that could not be confirmed by the discrete fou- component model. Because it can isolate autofluorescence as a separate component using the distinct lifetime profiles of endogenous fluorophores, uFLIM could provide a clearer pH signal by separating background from specific signal (Hwang et al. 2025). This would be particularly useful for

cellular imaging where autofluorescence from NADH, FAD and other endogenous fluorophores can contaminate the pH sensor signal (Datta et al. 2020).

Employing a hybrid initialisation strategy that combines the current model with the proposed approach could improve robustness (Boutsidis and Gallopoulos 2008). The guided analysis, using the two extreme pH references (4.5 and 7.5), could provide an initial estimate of the spatial distribution of protonated and deprotonated sensor populations. Using these estimates as the starting point, unsupervised FLIM or phasor S-FLIM analysis could then be applied. This would accelerate convergence, ensuring that the result closely approximates the actual pH value while allowing the algorithm the freedom to discover the nature of all components present in the cellular data, including potential intermediate states or unexpected contributions. A further improvement, suggested by the mismatch between *in vitro* and cellular conditions discussed above, would be to derive the reference decays from cells expressing a calibration probe such as the GPI-anchored RpHLuorin2 of Linders et al. (2022), measured in defined pH buffers.

The ultimate goal of the analysis is to obtain clear, undistorted, multiplexed and dynamic image readouts. To achieve this, we could co-express a FLIM pH sensor (RpHLuorin2) with a FRET-based biosensor for calcium, cAMP or kinase activity (Verma et al. 2023). This would extend the analysis beyond simply calculating pH values towards a multi-parameter cellular physiology. Independent pH maps and biochemical activity within a living cell could be computed simultaneously using a single excitation wavelength together with uFLIM-FRET or phasor S-FLIM, providing insight into coupled physiological processes, such as the relationship between Golgi pH and calcium signalling during secretion. The same framework could also address the questions that motivated this project: how distinct pH states are organised within organelles; how they are disrupted by pharmacological perturbation of the vacuolar H^+-ATPase (Forgac 2007) or of ER-Golgi trafficking (Klausner et al. 1992), both of which have been shown to alter organelle pH when measured by FILM (Linders et al. 2022); and what sequence of pH states a cargo protein experiences in transit, which could be followed with a synchronised release system such as RUSH (Boncompain et al. 2012; Linders et al. 2022).

4.6 Conclusion

This project implemented a guided unmixing FLIM methodology for intracellular pH mapping. Following evaluation, spatially coherent pH maps were generated that distinguish broad environmental states, supporting the key principles of lifetime-based pH sensing. Validation on recombinant calibration data gave histogram peaks within 0.2 pH units of the nominal values at pH 4.5 and 7.5, and the method was applied to cellular probe datasets, demonstrating its practical utility for mapping organelle pH under favourable acquisition conditions.

Through critical evaluation, we have identified the method's major drawbacks; its susceptibility to systematic error when reference dynamics do not match cellular conditions, its limited resolution of intermediate pH values due to SNR limits and the similarity of the reference decays, and its model rigidity when constrained to four components. The differences in results observed between cellular datasets, despite identical analysis procedures, highlight the importance of consistent experimental conditions and sufficient photon budget for reliable results. The photon budget constraint necessitated 4 x 4 spatial binning, representing a trade-off between SNR and spatial resolution that must be carefully managed.

These findings provide justification for evolving the methodology in FLIM data analysis. Integrating robust experimental methodologies and quantitative sensors with the analytical power of unsupervised computational frameworks, such as uFLIM, initialised with guided estimates and calibrated against in-cell references, could lead to more refined quantitative subcellular pH mapping. Such an approach would allow us to move beyond qualitative maps towards quantitative, dynamic and multi-parametric images of cellular physiology. This would help unlock deeper analytical insight in both health and disease, contributing to our understanding of fundamental cell biology and potentially suggesting strategies for combating diseases in which pH homeostasis is disrupted.

References

Amigo, J.M., Babamoradi, H. and Elcoroaristizabal, S. 2015. Hyperspectral image analysis. A tutorial. *Analytica Chimica Acta* 896, pp. 34–51. Available at: https://doi.org/10.1016/j.aca.2015.09.030.

Amigo, J.M., Martí, I. and Gowen, A. 2013. Hyperspectral imaging and chemometrics. In: *Data handling in science and technology*. pp. 343–370. Available at: https://doi.org/10.1016/b978-0-444-59528-7.00009-0.

Appenzeller-Herzog, C., Roche, A.-C., Nufer, O. and Hauri, H.-P. 2004. PH-induced conversion of the transport lectin ERGIC-53 triggers glycoprotein release. *Journal of Biological Chemistry* 279(13), pp. 12943–12950. Available at: https://doi.org/10.1074/jbc.m313245200.

Becker, W. 2012. Fluorescence lifetime imaging – techniques and applications. *Journal of Microscopy* 247(2), pp. 119–136. Available at: https://doi.org/10.1111/j.1365-2818.2012.03618.x.

Berezin, M.Y. and Achilefu, S. 2010. Fluorescence lifetime measurements and biological imaging. *Chemical Reviews* 110(5), pp. 2641–2684. Available at: https://doi.org/10.1021/cr900343z.

Boncompain, G. et al. 2012. Synchronization of secretory protein traffic in populations of cells. *Nature Methods* 9(5), pp. 493–498. Available at: https://doi.org/10.1038/nmeth.1928.

Boorman. 2019. *Multimodal Coherent Anti-Stokes Raman Scattering (CARS) microscopy for the detection of Isotope-Labelled molecules*. Cardiff University School of Biosciences. Available at: https://orca.cardiff.ac.uk/id/eprint/130291/1/DaleBoormanCorrectedThesisFinalVersion.pdf [Accessed: 11 February 2026].

Boutsidis, C. and Gallopoulos, E. 2008. SVD based initialization: A head start for nonnegative matrix factorization. *Pattern Recognition* 41(4), pp. 1350–1362. Available at: https://doi.org/10.1016/j.patcog.2007.09.010.

Bower, A.J., Li, J., Chaney, E.J., Marjanovic, M., Spillman, D.R. and Boppart, S.A. 2018. High-speed imaging of transient metabolic dynamics using two-photon fluorescence lifetime imaging microscopy. *Optica* 5(10), p. 1290. Available at: https://doi.org/10.1364/optica.5.001290.

Bräuer, P., Parker, J.L., Gerondopoulos, A., Zimmermann, I., Seeger, M.A., Barr, F.A. and Newstead, S. 2019. Structural basis for pH-dependent retrieval of ER proteins from the Golgi by the KDEL receptor. *Science* 363(6431), pp. 1103–1107. Available at: https://doi.org/10.1126/science.aaw2859.

Callenberg, C. et al. 2021. Super-resolution time-resolved imaging using computational sensor fusion. *Scientific Reports* 11(1), p. 1689. Available at: https://doi.org/10.1038/s41598-021-81159-x.

Casey, J.R., Grinstein, S. and Orlowski, J. 2009. Sensors and regulators of intracellular pH. *Nature Reviews Molecular Cell Biology* 11(1), pp. 50–61. Available at: https://doi.org/10.1038/nrm2820.

Coelho, L.P., Peng, T. and Murphy, R.F. 2010. Quantifying the distribution of probes between subcellular locations using unsupervised pattern unmixing. *Bioinformatics* 26(12), pp. i7–i12. Available at: https://doi.org/10.1093/bioinformatics/btq220.

Cotlet, M. et al. 2001. Excited-State dynamics in the enhanced green fluorescent protein mutant probed by Picosecond Time-Resolved Single Photon Counting spectroscopy. *The Journal of Physical Chemistry B* 105(21), pp. 4999–5006. Available at: https://doi.org/10.1021/jp003813i.

Crameri, F., Shephard, G.E. and Heron, P.J. 2020. The misuse of colour in science communication. *Nature Communications* 11(1), p. 5444. Available at: https://doi.org/10.1038/s41467-020-19160-7.

Datta, R., Heaster, T.M., Sharick, J.T., Gillette, A.A. and Skala, M.C. 2020. Fluorescence lifetime imaging microscopy: fundamentals and advances in instrumentation, analysis, and applications. *Journal of Biomedical Optics* 25(07), pp. 1–43. Available at: https://doi.org/10.1117/1.jbo.25.7.071203.

Digman, M.A., Caiolfa, V.R., Zamai, M. and Gratton, E. 2008. The Phasor Approach to Fluorescence Lifetime Imaging Analysis. *Biophysical Journal* 94(2), pp. L14–L16. Available at: https://doi.org/10.1529/biophysj.107.120154.

Forgac, M. 2007. Vacuolar ATPases: rotary proton pumps in physiology and pathophysiology. *Nature Reviews Molecular Cell Biology* 8(11), pp. 917–929. Available at: https://doi.org/10.1038/nrm2272.

Gabaldón, T. and Pittis, A.A. 2015. Origin and evolution of metabolic sub-cellular compartmentalization in eukaryotes. *Biochimie* 119, pp. 262–268. Available at: https://doi.org/10.1016/j.biochi.2015.03.021.

Gerritsen, H.C., Asselbergs, M. a. H., Agronskaia, A.V. and Van Sark, W.G.J.H.M. 2002. Fluorescence lifetime imaging in scanning microscopes: acquisition speed, photon economy and lifetime resolution. *Journal of Microscopy* 206(3), pp. 218–224. Available at: https://doi.org/10.1046/j.1365-2818.2002.01031.x.

Gowen, A.A. and Amigo, J.M. 2013. Applications of spectroscopy and chemical imaging in pharmaceutics. *Handbook for Biophotonics*, pp. 71–88. Available at: https://doi.org/10.1002/9783527643981.bphot083.

Gregor, I. and Patting, M. 2014. Pattern-Based linear unmixing for efficient and reliable analysis of multicomponent TCSPC data. In: *Springer series on fluorescence*. pp. 241–263. Available at: https://doi.org/10.1007/4243_2014_70.

Guo, J., Mai, F.-Y., Li, X.-Y., Liang, J.-R., Yang, W.-T. and Li, C.-G. 2025. Establishment of RpHluorin2-expressing cell and its application in monitoring JTC-801-induced alkaliptosis via multi-dimensional fluorescence detection approaches. *Frontiers in Cell and Developmental Biology* 13, p. 1727740. Available at: https://doi.org/10.3389/fcell.2025.1727740.

Heikal, A.A., Hess, S.T. and Webb, W.W. 2001. Multiphoton molecular spectroscopy and excited-state dynamics of enhanced green fluorescent protein (EGFP): acid–base specificity. *Chemical Physics* 274(1), pp. 37–55. Available at: https://doi.org/10.1016/s0301-0104(01)00486-4.

Hwang, W., McPartland, T., Jeong, S. and Evans, C.L. 2025. A robust method for autofluorescence-free immunofluorescence using high-speed fluorescence lifetime imaging

microscopy. *Scientific Reports* 15(1), p. 5503. Available at: https://doi.org/10.1038/s41598-025-89142-6.

Jain, A., Blum, C. and Subramaniam, V. 2009. Fluorescence Lifetime spectroscopy and imaging of visible fluorescent proteins. In: *Elsevier eBooks*. pp. 147–176. Available at: https://doi.org/10.1016/b978-0-444-53075-2.00004-6.

Jain, A. and Zoncu, R. 2022. Organelle transporters and inter-organelle communication as drivers of metabolic regulation and cellular homeostasis. *Molecular Metabolism* 60, p. 101481. Available at: https://doi.org/10.1016/j.molmet.2022.101481.

Keenan, M.R. and Kotula, P.G. 2004. Accounting for Poisson noise in the multivariate analysis of ToF-SIMS spectrum images. *Surface and Interface Analysis* 36(3), pp. 203–212. Available at: https://doi.org/10.1002/sia.1657.

Kervrann, C., Sorzano, C.O.S., Acton, S.T., Olivo-Marin, J.-C. and Unser, M. 2015. A guided tour of selected image processing and analysis methods for fluorescence and electron microscopy. *IEEE Journal of Selected Topics in Signal Processing* 10(1), pp. 6–30. Available at: https://doi.org/10.1109/jstsp.2015.2505402.

Khan, M.J., Khan, H.S., Yousaf, A., Khurshid, K. and Abbas, A. 2018. Modern Trends in Hyperspectral Image Analysis: A review. *IEEE Access* 6, pp. 14118–14129. Available at: https://doi.org/10.1109/access.2018.2812999.

Klausner, R.D., Donaldson, J.G. and Lippincott-Schwartz, J. 1992. Brefeldin A: insights into the control of membrane traffic and organelle structure. *The Journal of Cell Biology* 116(5), pp. 1071–1080. Available at: https://doi.org/10.1083/jcb.116.5.1071.

Köllner, M. and Wolfrum, J. 1992. How many photons are necessary for fluorescence-lifetime measurements? *Chemical Physics Letters* 200(1–2), pp. 199–204. Available at: https://doi.org/10.1016/0009-2614(92)87068-z.

Kornak, U. et al. 2007. Impaired glycosylation and cutis laxa caused by mutations in the vesicular H+-ATPase subunit ATP6V0A2. *Nature Genetics* 40(1), pp. 32–34. Available at: https://doi.org/10.1038/ng.2007.45.

Lee, D.D. and Seung, H.S. 1999. Learning the parts of objects by non-negative matrix factorization. *Nature* 401(6755), pp. 788–791. Available at: https://doi.org/10.1038/44565.

Linders, P.T.A., Ioannidis, M., Ter Beest, M. and Van Den Bogaart, G. 2022. Fluorescence Lifetime Imaging of pH along the Secretory Pathway. *ACS Chemical Biology* 17(1), pp. 240–251. Available at: https://doi.org/10.1021/acschembio.1c00907.

Linders, P.T.A., Peters, E., Ter Beest, M., Lefeber, D.J. and Van Den Bogaart, G. 2020. Sugary logistics gone wrong: membrane trafficking and congenital disorders of glycosylation. *International Journal of Molecular Sciences* 21(13), p. 4654. Available at: https://doi.org/10.3390/ijms21134654.

Liu, Y., Kim, H.-R. and Heikal, A.A. 2006. Structural basis of fluorescence fluctuation Dynamics of green fluorescent proteins in acidic environments. *The Journal of Physical Chemistry B* 110(47), pp. 24138–24146. Available at: https://doi.org/10.1021/jp062164t.

Mahon, M.J. 2011. pHluorin2: an enhanced, ratiometric, pH-sensitive green florescent protein. *Advances in Bioscience and Biotechnology* 02(03), pp. 132–137. Available at: https://doi.org/10.4236/abb.2011.23021.

Martinez, K.A., Kitko, R.D., Mershon, J.P., Adcox, H.E., Malek, K.A., Berkmen, M.B. and Slonczewski, J.L. 2012. Cytoplasmic pH Response to Acid Stress in Individual Cells of Escherichia coli and Bacillus subtilis Observed by Fluorescence Ratio Imaging Microscopy. *Applied and Environmental Microbiology* 78(10), pp. 3706–3714. Available at: https://doi.org/10.1128/aem.00354-12.

Masia, F., Dewitte, W., Borri, P. and Langbein, W. 2022. uFLIM — Unsupervised analysis of FLIM-FRET microscopy data. *Medical Image Analysis* 82, p. 102579. Available at: https://doi.org/10.1016/j.media.2022.102579.

Masia, F., Glen, A., Stephens, P., Borri, P. and Langbein, W. 2013. Quantitative chemical imaging and unsupervised analysis using hyperspectral coherent Anti-Stokes Raman Scattering microscopy. *Analytical Chemistry* 85(22), pp. 10820–10828. Available at: https://doi.org/10.1021/ac402303g.

MathWorks. [no date]. *Get started with hyperspectral image processing*. Available at: https://www.mathworks.com/help/images/get-started-with-hyperspectral-image-processing.html [Accessed: 11 February 2016].

Miesenböck, G., De Angelis, D.A. and Rothman, J.E. 1998. Visualizing secretion and synaptic transmission with pH-sensitive green fluorescent proteins. *Nature* 394(6689), pp. 192–195. Available at: https://doi.org/10.1038/28190.

Nasser, M. and Meller, A. 2021. Lifetime-based analysis of binary fluorophores mixtures in the low photon count limit. *iScience* 25(1), p. 103554. Available at: https://doi.org/10.1016/j.isci.2021.103554.

Paroutis, P., Touret, N. and Grinstein, S. 2004. The PH of the secretory Pathway: Measurement, determinants, and regulation. *Physiology* 19(4), pp. 207–215. Available at: https://doi.org/10.1152/physiol.00005.2004.

Petukh, M., Stefl, S. and Alexov, E. 2013. The role of protonation states in Ligand-Receptor recognition and binding. *Current Pharmaceutical Design* 19(23), pp. 4182–4190. Available at: https://doi.org/10.2174/1381612811319230004.

Ranjit, S., Malacrida, L., Jameson, D.M. and Gratton, E. 2018. Fit-free analysis of fluorescence lifetime imaging data using the phasor approach. *Nature Protocols* 13(9), pp. 1979–2004. Available at: https://doi.org/10.1038/s41596-018-0026-5.

Rennick, J.J., Nowell, C.J., Pouton, C.W. and Johnston, A.P.R. 2022. Resolving subcellular pH with a quantitative fluorescent lifetime biosensor. *Nature Communications* 13(1), p. 6023. Available at: https://doi.org/10.1038/s41467-022-33348-z.

Rivinoja, A., Hassinen, A., Kokkonen, N., Kauppila, A. and Kellokumpu, S. 2009. Elevated Golgi pH impairs terminal N-glycosylation by inducing mislocalization of Golgi glycosyltransferases. *Journal of Cellular Physiology* 220(1), pp. 144–154. Available at: https://doi.org/10.1002/jcp.21744.

Royle, S.J., Granseth, B., Odermatt, B., Derevier, A. and Lagnado, L. 2008. Imaging PHluorin-Based probes at hippocampal synapses. In: *Methods in molecular biology*. pp. 293–303. Available at: https://doi.org/10.1007/978-1-59745-261-8_22.

Salis, A. and Monduzzi, M. 2016. Not only pH. Specific buffer effects in biological systems. *Current Opinion in Colloid & Interface Science* 23, pp. 1–9. Available at: https://doi.org/10.1016/j.cocis.2016.04.004.

Schneider, C.A., Rasband, W.S. and Eliceiri, K.W. 2012. NIH Image to ImageJ: 25 years of image analysis. *Nature Methods* 9(7), pp. 671–675. Available at: https://doi.org/10.1038/nmeth.2089.

Schulte, A., Lorenzen, I., Böttcher, M. and Plieth, C. 2006. A novel fluorescent pH probe for expression in plants. *Plant Methods* 2(1), p. 7. Available at: https://doi.org/10.1186/1746-4811-2-7.

Scipioni, L., Rossetta, A., Tedeschi, G. and Gratton, E. 2021. Phasor S-FLIM: a new paradigm for fast and robust spectral fluorescence lifetime imaging. *Nature Methods* 18(5), pp. 542–550. Available at: https://doi.org/10.1038/s41592-021-01108-4.

Soltani, S., Paulson, J.G., Fong, E.J., Mumenthaler, S.M. and Armani, A.M. 2025. Denoising of fluorescence lifetime imaging data via principal component analysis. *Scientific Reports* 15(1), p. 40258. Available at: https://doi.org/10.1038/s41598-025-24022-7.

Torrado, B., Pannunzio, B., Malacrida, L. and Digman, M.A. 2024. Fluorescence lifetime imaging microscopy. *Nature Reviews Methods Primers* 4(1). Available at: https://doi.org/10.1038/s43586-024-00358-8.

Verma, A.K., Noumani, A., Yadav, A.K. and Solanki, P.R. 2023. FRET based biosensor: principle applications recent advances and challenges. *Diagnostics* 13(8), p. 1375. Available at: https://doi.org/10.3390/diagnostics13081375.

Warren, S.C. et al. 2013. Rapid global fitting of large fluorescence lifetime imaging microscopy datasets. *PLoS ONE* 8(8), p. e70687. Available at: https://doi.org/10.1371/journal.pone.0070687.

Wu, D. and Minton, A.P. 2015. Quantitative characterization of nonspecific self- and Hetero-Interactions of proteins in nonideal solutions via static light scattering. *The Journal of Physical Chemistry B* 119(5), pp. 1891–1898. Available at: https://doi.org/10.1021/jp510918d.

Wüstner, D. 2022. Dynamic mode decomposition of fluorescence loss in photobleaching microscopy data for Model-Free analysis of protein transport and aggregation in living cells. *Sensors* 22(13), p. 4731. Available at: https://doi.org/10.3390/s22134731.

Zhang, C., Wang, H., Huang, J. and Gao, Q. 2014. The visible to the near infrared narrow band acousto-optic tunable filter and the hyperspectral microscopic imaging on biomedicine study. *Journal of Optics* 16(12), p. 125303. Available at: https://doi.org/10.1088/2040-8978/16/12/125303.

Zhang, X., Zhang, X., Cheng, S., Fan, X., Bao, H., Zhou, S. and Ping, J. 2024. Spatiotemporal cell control via High-Precision electronic regulation of microenvironmental PH. *Nano Letters* 24(49), pp. 15645–15651. Available at: https://doi.org/10.1021/acs.nanolett.4c04174.